He 2021 mCRPC — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

11/12 samples (≥10 cells) intermix on one continuous manifold with no per-patient islands (radius-graph: 0 detached; iLISI median 2.11→5.56 of 11 samples). All three transcriptional lobes (213 / 298 / 322 cells) carry the trajectory.

Gate 2 — module geography → GO (non-branching)

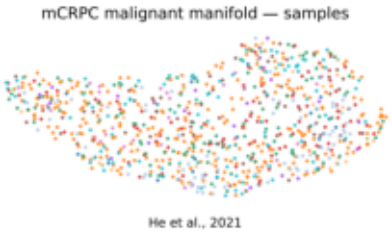
Single dominant Gast/pSom→Crest axis: gastrulation high at the origin declines steeply along pseudotime (ρ=−0.99); neural-crest anti-correlated (r=−0.82 vs Gast, −0.70 vs pSom), rising to a terminal readout (ρ=+0.85; crest-high decile mean pt=0.91). Gast↔pSom weakly positive (+0.18 — co-vary, not compete); joint pSom&Crest-high depleted (enrichment 0.15) — an inverse gradient, not a co-expression hub.

Gate 3 — cross-repression → NA

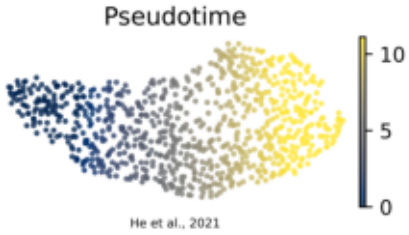
Single monotone axis — no bifurcation and no Mixed-type hub, so there is no branch pair to test for mutual exclusivity.

STEP FIGURES

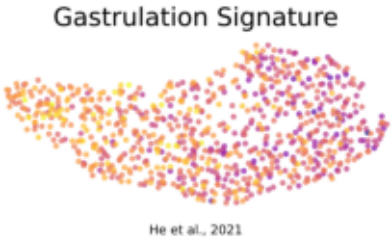
Manifold — samples



Pseudotime



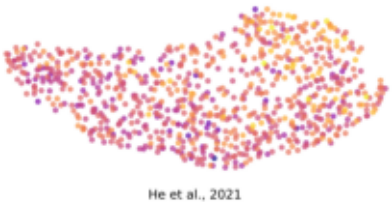
Gastrulation



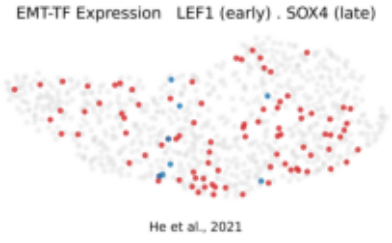
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Metastatic prostate cancer (He et al., 2021), SmartSeq2 — 15 samples / 2,170 cells (master scope); malignant only: 836 cells across 12 samples; 833 across 11 samples enter preprocessing (≥5-cell floor drops one 3-cell sample); the whole malignant manifold (no lobe scoping) enters the trajectory.

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. He's mCRPC malignant manifold is a single monotone gastrulation/somitogenesis → terminal-neural-crest gradient, matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. The axis spans all three transcriptional lobes (orthogonal to cell cycle; Crest–cyc ρ≈0.00, Gast–cyc ρ=−0.31), so the whole manifold is kept. Raw module means (Gast −0.30 / pSom +0.05 / Crest +0.05) place it in the Linear-pSom/Crest class; the gate turns on geography — a resolved single axis with a terminal Crest readout — not the means. EMT-TFs: canonical SNAI1/ZEB1/ZEB2/TWIST1 all below the 3% floor (FOXC/PRRX absent from the SmartSeq2 matrix), so the detectable EMT signal is the neural-crest/mesenchymal TF program rising to the terminus (SOX4 42%, ρ=+0.31); overlay LEF1 (early) → SOX4 (late), end-separation weak (Δpt=0.12).

Parameter choices

Min cells/sample	≥5 (geo-mean of 10→3 gap; 11/12)	**Integration**	Harmony θ=10; PCA50; spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=20, λ=0.3, μ=100
Root node	tip 0 = Gast/pSom-high, Crest-low	**Embedding**	n_pcs=15; lobes 213/298/322
Geometry	Linear (non-branching); no hub	**EMT-TFs**	LEF1 (early) → SOX4 (late); canonical set below 3% floor